

TKH Capstone Project



THE DATA ALCHEMISTS Cohort B Data Science



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Like most job hunters,
aspiring data scientists
are constantly looking
for guidance on how to
STAND OUT in the job
market.



We're constantly plagued with questions like:

- *Do I have enough skills?*
- *What skills do I need?*
- *How much should I expect to earn with my current skill set?*
- *Where can I go to learn more to make me a better candidate?*



As we neared the end of our fellowship with excitement mixed with doubt. We wondered where would we fit in the data science job market? That uncertainty sparked a solution: an AI partner that can serve individuals in the data science industry.



Let us introduce to you TKH Bootcamp Data Science AI Agent

Deploy ⋮

 AI Profiles

Select Persona

educator ▼

 Educator

Tone: Warm, Practical, Coach-Like

Goals:

- Bring real-world projects into curriculum
- Map outcomes to industry tools (cloud, Git, MLOps, advanced AI)

Style Rules:

- Use Checklists
- Include Rubrics
- Be Concise

TKH Bootcamp Data Science Dashboard

Interact with your assistant (Educator, Learner, Employer) and run built-in tools with an intuitive interface.

Chat with TKH Assistant

Use a Tool

Enter your message:

e.g., Give me a 2-week plan for Docker + CI basics.

Ask Persona



How WE make a difference?



Our tool will explore the data science industry through the scope of a student, teacher and employer. With collected data sets from job postings our AI agent will be able help each persona answer inquiries about the data science industry with prompts to assist with their day-to-day needs.



Questions That Guided Our Research

- *What are the most frequently requested skills in data science job postings?*
- *Which technical skills are associated with higher average salaries in data science job postings?*
- *How does the required education level correlate with salary?*
- *How do job postings for data analysts change over time?*





Research

Our real-world research focused on :

Jobs & Platforms:

Where are the jobs and what skills show up the most by region?

Levels & Trends:

How do expectations change as roles get more senior?

AI & Evolving Roles:

What happens to salary and role shape when AI shows up in the posting?

Skills Analysis:

Which skills and bundles of skills are most rewarded in this dataset?

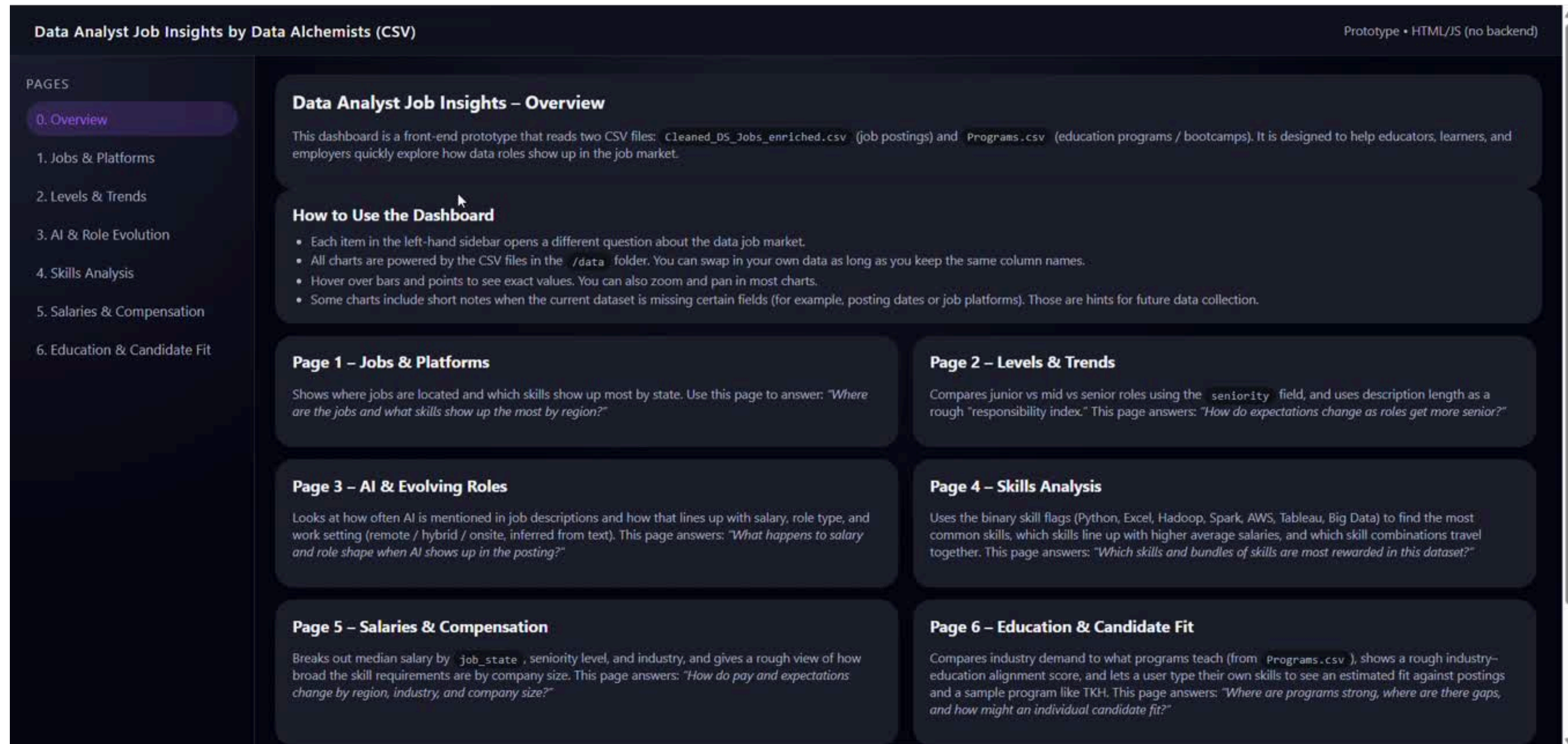
Salaries & Compensation:

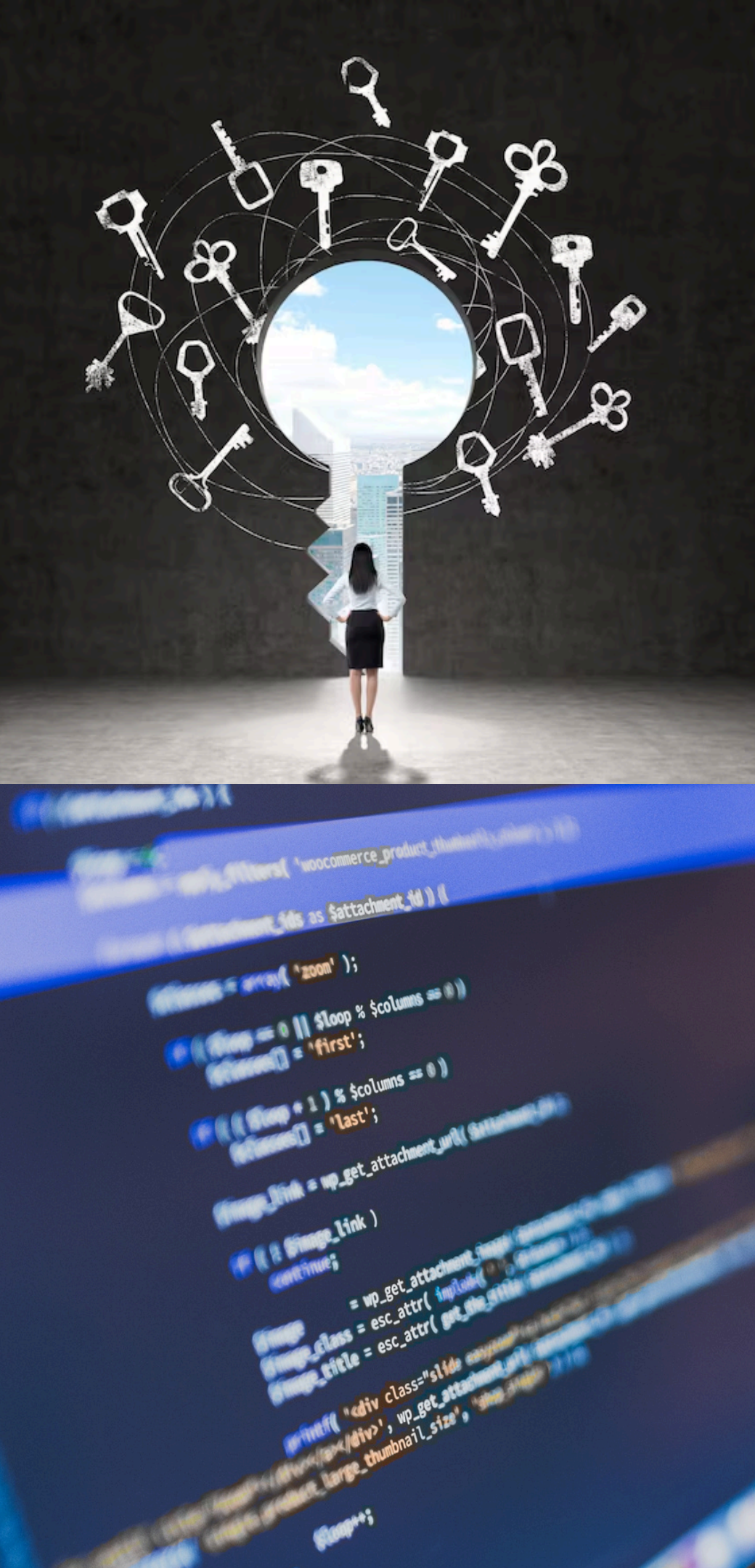
How do pay and expectations change by region, industry, and company size?

Education & Candidate Fit:

Where are programs strong, where are there gaps, and how might an individual candidate fit?

EDA Dashboard





Major Findings

- Python, SQL, AWS, and Tableau/Power BI remain top skills across roles and industries.
- Employers increasingly favor skills + experience over formal degrees.
- Salary varies sharply by industry (highest in Tech & Finance) and by location.
- AI-related postings show different compensation patterns and expectations.
- Programs often under-teach cloud, big data, and AI-related skills compared to industry demand.
- Large companies request a broader mix of technical skills, while smaller firms specialize more narrowly.

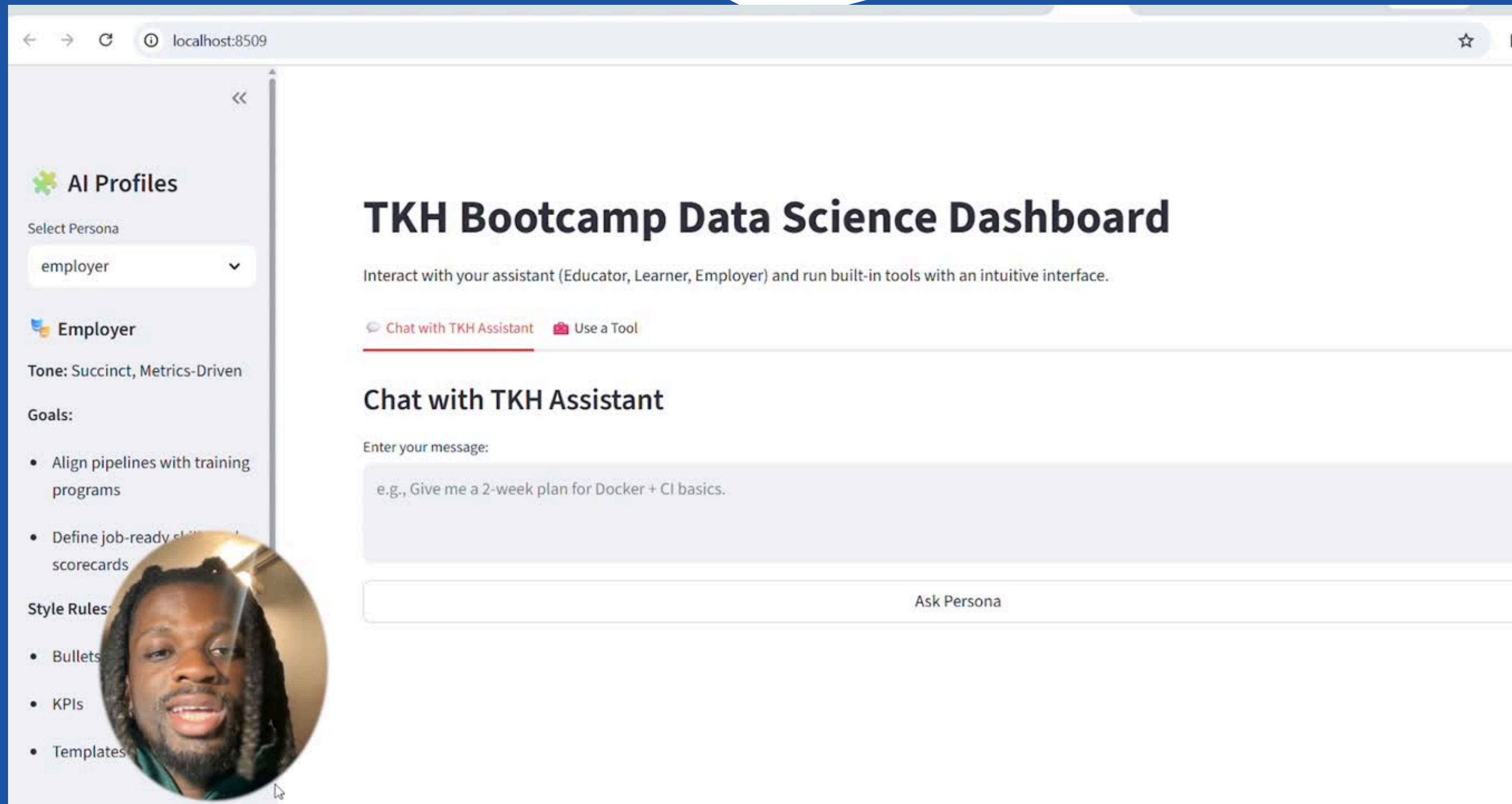


Personas

Our AI Agent takes the persona of 3 different users:

- **Learner** - The AI agent would work as a career coach.
- **Educator** - The AI agent can help schools or bootcamps make lesson plans that match job descriptions' skill requirements in the workforce.
- **Employer** - The AI agent can be a quick snapshot that shows where applicants and job requirements aren't in alignment.

Learner



The screenshot shows a web browser at localhost:8509 displaying the TKH Bootcamp Data Science Dashboard. The interface is divided into a left sidebar and a main content area. The sidebar, titled 'AI Profiles', includes a 'Select Persona' dropdown menu currently set to 'employer'. Below this, the 'Employer' profile is detailed with a 'Tone' of 'Succinct, Metrics-Driven' and a 'Goals' section containing two bullet points: 'Align pipelines with training programs' and 'Define job-ready skills with scorecards'. A 'Style Rules' section lists 'Bullets', 'KPIs', and 'Templates'. A circular profile picture of a man with dreadlocks is visible at the bottom of the sidebar. The main content area features the title 'TKH Bootcamp Data Science Dashboard' and a subtitle 'Interact with your assistant (Educator, Learner, Employer) and run built-in tools with an intuitive interface.' Below the subtitle are two buttons: 'Chat with TKH Assistant' (highlighted with a red underline) and 'Use a Tool'. The 'Chat with TKH Assistant' section includes a text input field with the placeholder 'Enter your message:' and a sample message 'e.g., Give me a 2-week plan for Docker + CI basics.' A 'Ask Persona' button is located at the bottom right of the chat area.

localhost:8509

TKH Bootcamp Data Science Dashboard

Interact with your assistant (Educator, Learner, Employer) and run built-in tools with an intuitive interface.

[Chat with TKH Assistant](#) [Use a Tool](#)

Chat with TKH Assistant

Enter your message:

e.g., Give me a 2-week plan for Docker + CI basics.

Ask Persona

AI Profiles

Select Persona

employer

Employer

Tone: Succinct, Metrics-Driven

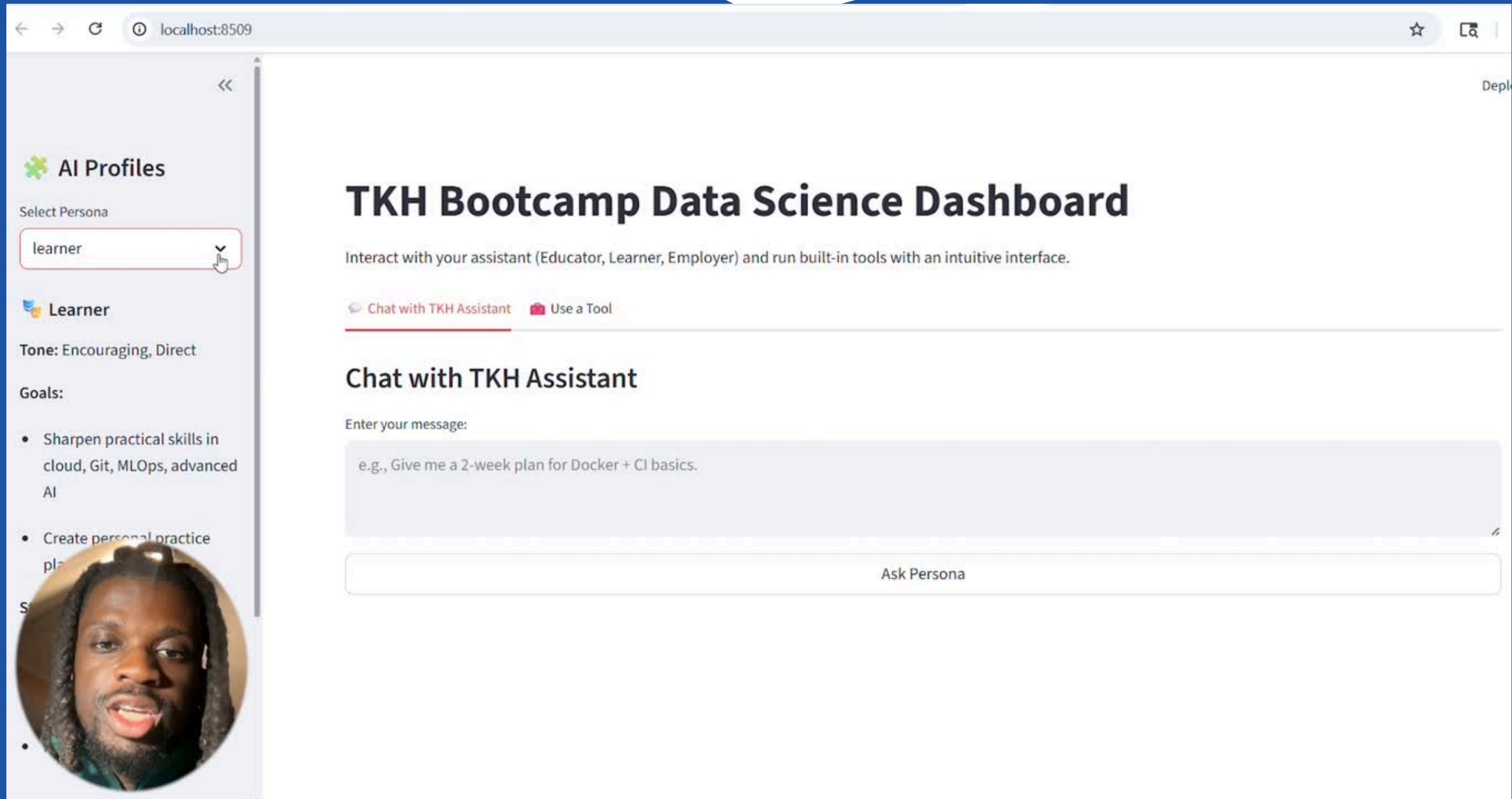
Goals:

- Align pipelines with training programs
- Define job-ready skills with scorecards

Style Rules

- Bullets
- KPIs
- Templates

Educator



The screenshot displays a web application interface for the TKH Bootcamp Data Science Dashboard. The browser's address bar shows the URL `localhost:8509`. On the left, a sidebar titled "AI Profiles" contains a "Select Persona" dropdown menu with "learner" selected. Below this, the "Learner" profile is detailed with a "Tone: Encouraging, Direct" and a "Goals" section listing tasks like "Sharpen practical skills in cloud, Git, MLOps, advanced AI" and "Create personal practice plan". A circular profile picture of a man is visible at the bottom of the sidebar. The main content area features the title "TKH Bootcamp Data Science Dashboard" and a subtitle "Interact with your assistant (Educator, Learner, Employer) and run built-in tools with an intuitive interface." It includes two buttons: "Chat with TKH Assistant" (highlighted with a red underline) and "Use a Tool". Below these is a section titled "Chat with TKH Assistant" with a text input field containing the placeholder text "e.g., Give me a 2-week plan for Docker + CI basics." and a button labeled "Ask Persona".

localhost:8509

AI Profiles

Select Persona

learner

Learner

Tone: Encouraging, Direct

Goals:

- Sharpen practical skills in cloud, Git, MLOps, advanced AI
- Create personal practice plan

TKH Bootcamp Data Science Dashboard

Interact with your assistant (Educator, Learner, Employer) and run built-in tools with an intuitive interface.

[Chat with TKH Assistant](#) [Use a Tool](#)

Chat with TKH Assistant

Enter your message:

e.g., Give me a 2-week plan for Docker + CI basics.

Ask Persona

Employer

The screenshot shows a web browser at localhost:8509 displaying the TKH Bootcamp Data Science Dashboard. The interface is divided into a left sidebar and a main content area.

Left Sidebar:

- AI Profiles** (with a green puzzle piece icon)
- Select Persona**: A dropdown menu currently showing "educator".
- Educator** (with a blue and orange icon)
- Tone:** Warm, Practical, Coach-Like
- Goals:**
 - Bring real-world projects into curriculum
 - M
 - Be Concise
- A circular profile picture of a man with dreadlocks.

Main Content Area:

- ## TKH Bootcamp Data Science Dashboard
- Interact with your assistant (Educator, Learner, Employer) and run built-in tools with an intuitive interface.
- Two buttons: "Chat with TKH Assistant" (with a speech bubble icon) and "Use a Tool" (with a toolbox icon).
- ### Chat with TKH Assistant
- Enter your message:
- A large text input field containing the placeholder text: "e.g., Give me a 2-week plan for Docker + CI basics."
- An "Ask Persona" button at the bottom of the input area.

At the top right of the main content area, there is a "Deploy" button and a menu icon.

Capstone Overview

Accomplishments

- A multi-page **HTML dashboard** visualizing key insights
- A **custom multi-persona AI Agent** executed via **Streamlit**
- Created a robust **GitHub repository**
- **6 structured workbooks** for **targeted EDA**
- A **Kanban board** to manage weekly milestone requirements

Roadblocks

- Incomplete or **inconsistent data** for EDA and RAGs
- Integrating the **API Key, AI agent and RAG data pipeline**
- Handling duplicates and **code formatting issues**
- Understanding and utilize Git commands
- **Power BI modeling errors** and therefore pivoting to HTML pages
- Managing team dynamics and missing teammates

Next Steps

- Add a **chat history** per persona and **implement a memory system**
- Implement a **job board** that streamlines up to date postings
- **Additional user testing** to address any errors for official deployment

Lessons Learned

- Developed a project concept and designed a dashboards that tells a cohesive story
- Demonstrated the use of key skills learned throughout the TKH Bootcamp
- Showcased the implementation of real-world datasets into the machine learning tools
- Created AI tools that can be used can streamline analysis and enhance learning for future students

Acknowledgments

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- Anil, Maurice & Farukh
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- Cohort B Data Science Peers
- Everyone who contributed feedback, resources, and support during this project





GIT HUB LINK

<https://github.com/nyckienite/TKH-Capstone-Project.git>

